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DEMOGRAPHIC HISTORY OF THE WILD FIELDS AND BUDJAK (10000BC - 1400AD)

A Regional Literature Review of Population and Settlement (10000BC - 1400AD).

10 November 2026 - Aust Kätzchen | austkaeztchen@confoederatio.org

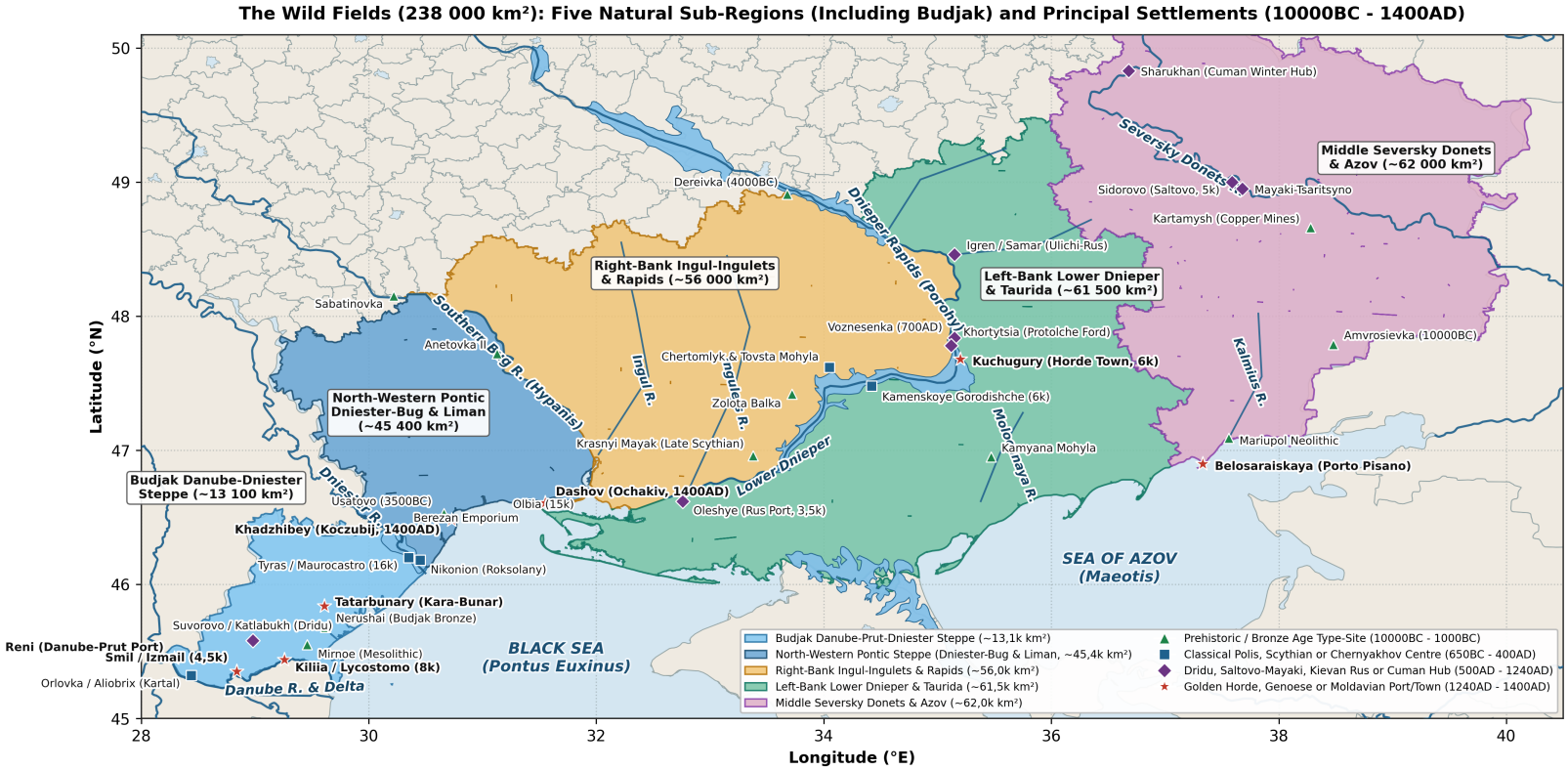


Figure 1. Cartographic overview of the Wild Fields and Budjak showing the five natural hydrological sub-regions and principal settlements (10000BC - 1400AD).

Geographical Definition, Ecological Regimes, and Five-Part Natural Sub-Regional Partition

Spatial Boundaries and Natural Steppe Hydrological Corridors

The analytical region designated as the Wild Fields (historical Polish *Dzikie Pola*, Latin *Loca Deserta*, Ukrainian *Dyke Pole*, and the western wing of *Desht-i Kipchak*) encompasses the entire southern Ukrainian Pontic-Azov steppe zone north of the Perekop Isthmus and south of the forest-steppe ecotone, spanning approximately 238 000 km^2 (4 011 5-arcminute grid cells covering 231 135 km^2 of land and estuarine cell area) between $45{,}2°$N and $49{,}9°$N latitude and $28{,}2°$E and $40{,}2°$E longitude [1][2]. This literature review and its companion dasymetric calibration cover the period from the Pleistocene-Holocene transition (10000BC) to the late Golden Horde, Genoese, Moldavian, and Lithuanian benchmark of 1400AD;

post-1400AD developments (the Ottoman-Tatar frontier, Zaporozhian *Siches*, and Imperial/Modern colonisation) require separate regime handling. The Crimean Peninsula south of Perekop and Chongar (~26 945 km^2) is treated in the companion Crimean synthesis, while the forest-steppe principalities of Galicia-Volhynia, Kiev-Pereyaslavl, and Chernigov-Seversk bound the Wild Fields along the natural forest-steppe line to the north.

Across the Holocene (10000BC - 1400AD), demographic carrying capacity in the Wild Fields was governed by the contrast between two ecological zones. The first zone comprised the open interfluvial feather-grass steppe (*plakor*), consisting of gently undulating loess plateaus covered by *Stipa-Festuca* chernozems and southern chestnut soils receiving 350 - 480 mm of annual precipitation; prior to modern deep well-drilling, these interfluvial watersheds lacked permanent summer surface water and functioned primarily as meridian grazing corridors for mobile pastoralists [1][3]. The second zone comprised the incised river valleys, rapids, floodplain forests (*plavni*), freshwater lakes, and coastal estuaries (*limany*). Here, the great southward-flowing rivers, including the Lower Danube (Chilia/Kiliia arm) and its Budjak lakes (Yalpug, Kugurlui, Katlabukh, Kitai, Kagul), the Dniester (*Tyras*), Tiligul, Southern Bug (*Hypanis*), Ingul, Ingulets, Lower Dnieper (*Borysthenes*) with its nine granite rapids (*Porohy*) and the *Veliky Lug* ("Great Meadow") gallery forest, Samara, Orel, Konka, Molochnaya, Kalmius, Mius, and Middle Severksy Donets, carved sheltered micro-ecologies rich in gallery timber, fresh water, anadromous sturgeon and cyprinid fisheries, winter reed-bed fodder, and defensive bluffs [4][5][6][7].

Five-Part Natural Sub-Regional Framework (Including Budjak)

To ensure that both the dasymetric areal mask and vector cartography (Figure 1 and Table 1) follow strict natural hydrological and ecological boundaries while treating Budjak as a distinct historical-demographic region, the 238 000 km^2 domain is partitioned into five natural sub-regions.

Sub-region I encompasses Budjak (Danube-Prut-Dniester Steppe and Liman, ~13 100 km^2; 227 grid cells), bounded naturally by the Lower Danube River and Chilia Delta on the south, the Prut confluence and Moldovan border on the west, the Black Sea coast (with the Sasyk, Shagany, Alibey, and Burnas lagoons) on the south-east, and the Dniester River and Dniester Liman on the north-east. Unlike the interior continental steppe, Budjak combines an open interior *plakor* (drained by the Kogilnik, Chaga, Sarata, and Khadzhider rivers) with two high-carrying-capacity maritime-fluvial interfaces: the Danube floodplain lakes (Kagul, Yalpug, Katlabukh, Kitai) and the western bank of the Dniester Liman. Consequently, Budjak served continuously as the primary demographic bridge between the Balkan-Danubian world (Gumelnita, Getae, Roman Lower Moesia, First and Second Bulgarian Empire *Dridu* culture, Byzantium, Genoese *Lycostomo*, and Moldavia) and the Pontic steppe (Yamnaya/Nerushai, Scythians, Sarmatians, Pechenegs, Cumans, and Golden Horde *Aq Qerman*) [8][9][10][11].

Sub-region II spans the North-Western Pontic Steppe (Dniester-Bug Interfluve and Tiligul-Berezan-Bug Liman, ~45 400 km^2; 475 grid cells), bounded naturally on the south-west by the Dniester River and Liman, on the north by the Kodyma River and forest-steppe ecotone, on the north-east by the Southern Bug River (*Hypanis*) and Bug Liman, and on the south by the Black Sea coast. Historically, it formed the core of the Eneolithic Usatovo complex (Usatovo, Mayaki), the Milesian Greek *poleis* and *chora* networks of Olbia, Berezan, Nikonion, and Koshary, dense Chernyakhov coastal villages (Koblevo, Kamenka-Anchekrak), Tivertsi and Ulichi forest-steppe frontier settlements, and the late fourteenth-century ports of Khadzhibey (*Koczubij*) and Dashov (*Ochakiv*) [12][13][14][15].

Sub-region III covers the Right-Bank Bug-Dnieper Interfluve, Ingul-Ingulets Uplands, and Dnieper Rapids Right Bank (~56 000 km^2; 922 grid cells), bounded naturally on the west by the Southern Bug River, on the north by the forest-steppe ecotone along the Dnieper Upland, and on the east and south-east by the Dnieper River (*Borysthenes*) and Dnieper-Bug Estuary. Encompassing the Syniukha, Ingul, Ingulets, Visun,

Saksahan, and Bazavluk basins, this sub-region hosted the Dereivka Eneolithic enclosure, the Sabatinovka and Ingul Catacomb Bronze Age heartland, the Royal Scythian giant kurgans (Chertomlyk, Tovsta Mohyla, Alexandropol), and the chain of Late Scythian right-bank stone fortresses (Zolota Balka, Gavrilovka, Krasny Mayak, Konsulovskoye, Nikolaevka-Kazatskoye) [5][16][17].

Sub-region IV comprises the Left-Bank Lower Dnieper, Veliky Lug, Samara-Orel, and Northern Taurida Steppe (~61 500 km^2; 1 107 grid cells), bounded on the west by the Dnieper River channel, on the north by the Orel (*Oril*) River and forest-steppe ecotone, on the east by the Dnieper-Donets and Azov Upland watershed (Berda River divide), and on the south by the Black Sea, Perekop-Sivash, and Sea of Azov. It contained the Vasylivka Mesolithic cemeteries, the Kamyana Mohyla sanctuary, the Mikhailovka Yamnaya stone citadel, the 1 200 ha Scythian city of Kamenskoye Gorodishche, the Voznesenka princely camp, the Kievan Rus ports of Oleshye, Protolche (Khortytsia), and Igren-Samar, and the Golden Horde cities of Velyki Kuchugury, Mamai-Surka, and Konski Vody [18][19][20][21].

Sub-region V encompasses the Middle Seversky Donets, Donets Ridge, and Kalmius-Mius Azov Basin (~62 000 km^2; 1 280 grid cells), bounded on the west by the Dnieper-Donets and Azov Upland watershed, on the north-west by the forest-steppe ecotone, on the east by the Aidar-Derkul-Donets-Mius basin frontier, and on the south by the Sea of Azov. Spanning the chalk bluffs of the Middle Seversky Donets and Oskol valleys, the elevated Donets Ridge (*Donetskiy Kryazh*), and the Kalmius-Mius coastal slope, this eastern sector supported the Amvrosievka Palaeolithic camp, the Mariupol Neolithic cemetery, the Bronze Age copper mines of Kartamysh, the eighth- to tenth-century Khazar-Alanic-Bulgar urban network of the *Saltovo-Mayaki* culture (Sidorovo, Mayaki-Tsaritsyno, Sukhaya Gomolsha, Mokhnach), the Polovtsian winter capitals of Sharukhan, Sugrov, and Balin, and the Golden Horde and Italian Azov port at Belosaraiskaya (*Porto Pisano*) [6][22][23].

Table 1. Natural hydrological and ecological partition of the Wild Fields and Budjak into five analytical sub-regions (10000BC - 1400AD).

Sub-Region	Area (% Share)	Natural Hydrological & Ecological Boundaries	Representative Pre-1400AD Anchor Centres
I. Budjak (Danube-Prut-Dniester Steppe & Liman)	13 100 (5,5%)	Danube R. & Delta (S), Prut/Moldova (W), Black Sea (SE), Dniester R. & Liman (NE)	Mirnoe, Nerushai, Orlovka (Aliobrix), Tyras / Aq Qerman (Maurocastro), Kiliia (Lycostomo), Smil (Izmail), Reni
II. NW Pontic Steppe (Dniester-Bug Interfluve & Liman)	45 400 (19,1%)	Dniester R. (SW), Kodyma & Forest-Steppe (N), Southern Bug R. & Liman (NE)	Usatovo, Mayaki, Nikonion, Berezan, Olbia, Koshary, Sabatinovka, Anetovka II, Khadzhibey, Dashov (Ochakiv)
III. Right-Bank Bug-Dnieper, Ingul-Ingulets & Rapids	56 000 (23,5%)	Southern Bug R. (W), Forest-Steppe Ecotone (N), Dnieper R. & Estuary (E/SE)	Dereivka, Chertomlyk, Tovsta Mohyla, Zolota Balka, Gavrilovka, Krasny Mayak, Bashmachka, Voznesensk
IV. Left-Bank Lower Dnieper, Veliky Lug & N. Taurida	61 500 (25,8%)	Dnieper R. (W), Orel R. & Forest-Steppe (N), Azov Upland Divide (E), Perekop/Azov (S)	Vasylivka, Mikhailovka, Kamyana Mohyla, Kamenskoye, Voznesenka, Khortytsia (Protolche), Oleshye, Kuchugury
V. Middle Seversky Donets, Donets Ridge & Azov Basin	62 000 (26,1%)	Dnieper-Donets Divide (W), Forest-Steppe (NW), Aidar-Mius Frontier (E), Azov Sea (S)	Amvrosievka, Mariupol Neolithic, Kartamysh, Sidorovo, Mayaki-Tsaritsyno, Sukhaya Gomolsha, Sharukhan, Sugrov
Total Wild Fields (10000BC - 1400AD)	238 000 (100,0%)	Five Natural Hydrological & Ecological Sub-Regions (4 011 Grid Cells)	191 Georeferenced Pre-1400AD Settlements in Regional Gazetteer

Prehistoric Foragers, Eneolithic Horse-Keepers, and Bronze Age Agro-Pastoralists (10000BC - 1000BC)

Epigravettian, Mesolithic, and Neolithic Fluvial Refugia (10000BC - 4500BC)

At the Pleistocene-Holocene transition (10000BC), the retreat of periglacial tundra-steppe and the drowning of the exposed north-western Black Sea shelf into brackish *liman* estuaries concentrated Upper Palaeolithic Epigravettian and Early Mesolithic bands along limestone canyon gorges, the Budjak coastal rivers, and granite river rapids [4][14]. Archaeological surveys synthesised by Stanko and Olenkovskiy across the Southern Bug (Anetovka II, 47,72°N, 31,13°E in Sub-region II), the Krynka-Mius basin (Amvrosievka bison kill site and bone bed, 47,79°N, 38,48°E in Sub-region V), and the Lower Dnieper document specialised communal hunting of steppe bison (*Bison priscus*), wild horse (*Equus ferus*), and saiga antelope [4]. During the Kukrek and Grebeniki Mesolithic horizons (8000BC), large base camps such as Mirnoe (45,58°N, 29,46°E in Sub-region I: Budjak), the sandstone rock-shelter sanctuary of Kamyana Mohyla (46,95°N, 35,47°E in Sub-region IV) on the Molochnaya River, and the communal cemeteries along the Dnieper Rapids (Vasylivka I-III, Voloske, Igren VIII, containing 30 - 50+ burials per necropolis) demonstrate incipient territoriality around high-yield fishing weirs. At 10000BC, total regional population across 238 000 km^2 is estimated at 3,8k (0,30k in Budjak [I], 0,80k in NW Pontic [II], 0,95k in Right-Bank [III], 0,95k in Left-Bank [IV], and 0,80k in Middle Donets [V]), rising to 6,8k by 8000BC.

During the Neolithic Atlantic climatic optimum (6000BC - 5000BC), four ceramic-using traditions established riverine and estuarine villages across the five sub-regions. Along the shores of Lakes Yalpug, Kagul, and Katlabukh in Budjak (Sub-region I, 6000 - 4500BC), early agro-foraging communities interacted with Balkan Starcevo-Cris and Linearbandkeramik groups, lifting Budjak's population from 1,1k at 6000BC to 1,9k at 5000BC [9][14]. In Sub-regions II and III (6200 - 5000BC), the Bug-Dniester Culture concentrated along the Southern Bug rapids between Savran, Migeya, and Bazkov Ostrov (48,03°N, 30,96°E), combining impressed pottery, flint sickle blades, and early domesticates [14]. Across Sub-regions III and IV (6000 - 4500BC), the Sursko-Dnieper and Dnieper-Donets II complexes were anchored on the islands and banks of the Dnieper Rapids (Surskoy Island, Nikolskoye, Vovnigi with 130+ skeletons) and the Samara-Orel confluence [24]. In Sub-region V (5500 - 4500BC), the Lower Don-Azov and Mariupol horizon was exemplified by the 122-burial red-ochre corridor cemetery at Mariupol (47,09°N, 37,56°E) at the mouth of the Kalmius River. Total Wild Fields population doubled from 14,0k at 6000BC to 24,0k at 5000BC.

Bolgrad-Aldeni (Gumelnita), Sredny Stog, and Usatovo Maritime Chiefdoms (4500BC - 3000BC)

Between 4500BC and 3000BC, the spread of Balkan-Carpathian copper metallurgy and steppe horse management reshaped the region, with Budjak (Sub-region I) and the Dniester-Bug corridor (Sub-region II) acting as the primary demographic engines [9][25][26]. In Sub-region I (4500 - 3000BC), as excavated by Subbotin around Lakes Yalpug, Kagul, Katlabukh, and Kitai (Bolhrad, Ozernoye, Etulia VI, Plavni, Nagornoye, Suvorovo, Chervonyi Yar, and Orlovka/Kartal), sedentary Balkan Chalcolithic farmers of the Bolgrad-Aldeni variant of the Gumelnita Culture established dense tell-like villages of 20 - 40 wattle-and-daub houses in south-western Budjak (4200 - 3800BC, 3,8k in Sub-region I at 4000BC) [9][27]. In the mid-fourth millennium (3500 - 3000BC), interaction between late Cucuteni-Trypillia CII, Cernavoda I, and the Budjak Culture (*Budzhakskaya kultura*, represented by thousands of Yamnaya/Usatovo kurgan burials at Nerushai, Semenovka, Mayaki, Trapovka, Borisovka, Sarata, and Tarutyne) raised Budjak's population to 9,5k at 3500BC and 8,5k at 3000BC [1][9].

East of the Dniester Liman in Sub-region II (3500 - 3000BC) around the Khadjibey and Kuialnyk estuaries, fortified limestone-built centres of the Usatovo complex at Usatovo (46,53°N, 30,66°E) and Mayaki

(46,41°N, 30,27°E) presided over 14,5k inhabitants at 3500BC [14]. Combined with Budjak (9,5k), the western maritime steppes (Sub-regions I and II) held 24,0k inhabitants (35,3% of the Wild Fields total). Along the Dnieper Rapids and Lower Dnieper in Sub-regions III, IV, and V (4000 - 3000BC), Sredny Stog horse-keeping communities at Dereivka (48,91°N, 33,68°E, Sub-region III) and the three-phase stone-walled Yamnaya stronghold at Mikhailovka (47,29°N, 33,95°E, Sub-region IV) anchored expanding wagon-pastoral lineages across the Ingulets, Molochnaya, and Seversky Donets, lifting total regional population from 37,0k at 4000BC to 68,0k at 3500BC and 75,0k at 3000BC [24][26].

Catacomb, Babino, and the Sabatinovka-Srubnaya Sedentary Boom (3000BC - 1000BC)

During the Middle Bronze Age Catacomb (*Ingul* and *Donets* variants, 2800 - 2000BC) and Babino (*Mnogovalikovaya*, 2200 - 1800BC) horizons, pastoral density rose steadily to 95,0k at 2000BC (8,0k in Budjak [I], 20,0k in NW Pontic [II], 24,5k in Right-Bank [III], 24,0k in Left-Bank [IV], and 18,5k in Middle Donets [V]) [1][9].

The prehistoric Bronze Age culminated during the Late Bronze Age Sabatinovka-Noua (1600 - 1200BC, Sub-regions I-IV), Srubnaya (1600 - 1100BC, Sub-regions IV-V), and Belozerka (1200 - 950BC) horizons [16][28][29]. In Budjak (Sub-region I), over 200 Late Bronze Age Sabatinovka-Coslogeni-Noua and Belozerka (Tudora-type) settlements lined the Kogilnik, Sarata, Alkaliya, and Danube lake shores (Zhovtneve/Karakurt, Ogorodnoye, Suvorovo, Tatarbunary, Diviziya, Tuzly, Shabo, and Tudora), supporting 15,0k inhabitants at 1000BC [9][29]. In Sub-region II (Dniester-Bug Interfluve, 33,0k at 1000BC), multi-roomed limestone-foundation villages such as Sabatinovka (48,15°N, 30,22°E), Voronovka II, and Tashlyk practised intensive cattle-and-millet mixed farming [16]. Across the Right-Bank Ingul-Ingulets (Sub-region III, 36,0k), Left-Bank Lower Dnieper (Sub-region IV, 34,0k), and Donets-Azov Copper Province (Sub-region V, 27,0k), hundreds of stone-built villages (Zmiivka, Novokievka, Staryi Krym Ingul) flourished alongside the Srubnaya copper-mining and bronze-casting complexes of the Bakhmut basin (Kartamysh, Usovo Ozero, Kapitanovo in Sub-region V), bringing total Wild Fields population to 145,0k at 1000BC [3][28].

Classical Greek Poleis, Royal Scythia, Late Scythian Fortresses, and the Chernyakhov Horizon (700BC - 400AD)

Greek Poleis and Chora Networks in Budjak (Sub-Region I) and the Dniester-Bug Liman (Sub-Region II)

Beginning in the mid-seventh century BC (~647BC), Milesian Greek colonisation established dense urban and rural *chora* networks across both Sub-region I (Budjak) and Sub-region II (Dniester-Bug Interfluve) [8][12][30]. In Sub-region I (Budjak, supporting 27,0k at 500BC, 18,0k at 0AD, and 29,0k at 300AD), Milesian colonists on the rocky western promontory of the Dniester Liman (modern Bilhorod-Dnistrovskyi) founded Tyras (~600BC), which grew into a fortified, coin-minting *polis* of 6,0k inhabitants at 500BC - 300BC, surrounded by a dense western Dniester Liman *chora* (Shabo, Zatoka, Mologa II, Pivdenne I, Belenkoye) [8][31]. At the southwestern corner of Budjak overlooking the Danube crossing at the Kartal narrows (near Reni), the fortified Geto-Scythian and later Roman-era emporium of Orlovka/Kartal (ancient Aliobrix, 45,32°N, 28,44°E, accompanied by Novoselskoye and Reni) controlled riverine trade between the Danube Delta, the Prut, and the interior Budjak steppe (1,5k - 2,2k inhabitants, where gold staters of Cyzicus and later Roman legionary bricks of *Legio V Macedonica* and *Classis Flavia Moesica* are attested) [8][32].

In Sub-region II (Dniester-Bug Interfluve, supporting 58,0k at 500BC, 40,0k at 0AD, and 59,0k at 300AD), Berezan (~647BC, 2,5k) and Pontic Olbia (~600BC, 50 - 55 ha walled upper and lower city with 12,0k - 15,0k

urban residents) on the Bug-Dnieper Liman anchored over 150 rural *chora* villages (*Shirokaya Balka*, *Kozyrka*, *Chertovatoe VII*, *Viktorovka I*, *Staraya Bogdanovka II*, *Didova Khata*) housing 20,0k - 25,0k Greco-Scythian grain cultivators [12][13][33]. On the eastern bank of the Dniester Liman opposite Tyras, Nikonion (Roksolany, 3,0k at 500 - 400BC) and its *chora* (Nadlimanskoye) linked the Dniester trade to Koshary (Odessos, 1,4k at the Tiligul mouth) and the Odessa Bay settlements (Zhevakhova Gora, Luzanovka) [15][34].

Royal Scythia, Kamenskoye Gorodishche, Late Scythian Fortresses, and the Chernyakhov Peak (500BC - 400AD)

In the interior steppes of Sub-regions III, IV, and V, fifth- and fourth-century BC Royal Scythia reached its demographic zenith along the Lower Dnieper Rapids and *Veliky Lug* (*Gerrhus*) [17][18]. At Kamenskoye Gorodishche (47,48°N, 34,42°E in Sub-region IV, enclosing 1 200 ha with its 32 ha Znamenka acropolis and the 800-burial Mamai-Hora necropolis), a resident metallurgical and administrative population of 6,0k (plus 3,5k - 4,2k at Znamenka) operated alongside the Royal Scythian Giant Kurgans of Sub-region III (Chertomlyk, Tovsta Mohyla, Alexandropol) and Sub-region IV (Solokha, Melitopol, Gaimanova Mohyla) [17][18]. In Sub-region V (Middle Seversky Donets and Azov, 36,0k at 500BC), Scythian forest-steppe hillforts (Velikaya Gomolsha, Koropovye Khutora) and Azov coastal camps linked the Donets corridor to the Royal Scythians. Total Wild Fields population reached a classical peak of 235,0k at 500BC (27,0k in Budjak [I], 58,0k in NW Pontic [II], 56,0k in Right-Bank [III], 58,0k in Left-Bank [IV], and 36,0k in Middle Donets [V]).

Following the third-century BC Sarmatian advance (which lowered regional population to 165,0k at 0AD), sedentary agriculture rebounded between 250BC and 200AD along both banks of the Lower Dnieper gorge (Sub-regions III and IV) via 16 bastioned stone fortresses (Zolota Balka, Krasny Mayak, Gavrilovka, Konsulovskoye, Nikolaevka-Kazatskoye, Lvove, Sabinka, and Hannivka in Sub-region III; Lyubimovka, Znamenskoye, Gornostaevka, and Kairy in Sub-region IV) [5][35][36][37]. Simultaneously, Roman vexillations stationed at Tyras and Orlovka/Aliobrix (Sub-region I) and Olbia (Sub-region II) stabilised the coastal zone, lifting regional population to 205,0k at 200AD [38].

Under the multi-ethnic Chernyakhov Culture (250 - 375AD), hundreds of stone-built and timber villages covered Budjak (29,0k in Sub-region I at Bolhrad-Yalpug, Etulia, Chervonyi Yar, Kubei, Nagornoye, Belenkoye, and Mologa II), the Dniester-Bug Liman (59,0k in Sub-region II at Koblevo, Kamenka-Anchekrak, and Ranconia), the Right-Bank Ingul-Ingulets and Rapids (62,0k in Sub-region III at Bashmachka and Snihurivka), the Left-Bank Lower Dnieper (54,0k in Sub-region IV), and the Donets corridor (36,0k in Sub-region V), reaching an antique high-water mark of 240,0k at 300AD before the Hunnic invasion of 375AD reduced the Wild Fields to 150,0k at 400AD (18,0k in I, 36,0k in II, 38,0k in III, 35,0k in IV, and 23,0k in V) [32][39][40].

Early Medieval Nomads, the Balkan-Danubian Dridu Culture in Budjak, the Saltovo-Mayaki Boom, and the Cuman-Rus Condominium (400AD - 1240AD)

Antae/Penkovka Rapids Refugia, Voznesenka, and the Early Medieval Trough (400AD - 750AD)

Between 400AD and 700AD, as Kutrigur and Utigur Bulgars, Avars, and Old Great Bulgaria under Khan Kubrat traversed the open steppes, regional population bottomed out at 125,0k in 500AD (14,0k in Budjak [I], 28,0k in II, 34,0k in III, 29,0k in IV, 20,0k in V) before recovering to 135,0k in 600AD and 160,0k in 700AD [40][41]. Sedentary life persisted along the Lower Danube and Dniester crossings of Budjak (Orlovka/Kartal, Novoselskoye) and the granite islands of the Dnieper Rapids (Voloshskoye, Igren-Samar, Balka Kantsyrka, and the c. 700AD princely memorial camp at Voznesenka in Sub-region IV) [41].

The Balkan-Danubian (Dridu) Boom in Budjak (Sub-Region I) and the Saltovo-Mayaki Urban Surge on the Donets (Sub-Region V) (750AD - 965AD)

During the eighth through tenth centuries (750 - 965AD), the two opposite wings of the Wild Fields, Sub-region I (Budjak) in the south-west and Sub-region V (Middle Seversky Donets) in the north-east, experienced parallel sedentary demographic booms. Following Asparukh's crossing of the *Onglos* in southern Budjak (680AD) and the consolidation of the First Bulgarian Empire north of the Danube, archaeological surveys around Lakes Yalpug, Kugurlui, Katlabukh, Kitai, and Kagul have documented over 70 eighth- to tenth-century sedentary Balkan-Danubian (*Dridu* or *Pliska-Preslav*) villages (Suvorovo I-VI, Bogatoye, Ogorodnoye, Kholmskoye, Chervonyi Yar, Plavni, Etulia, Nagornoye, Safyany/Izmail, and Orlovka/Kartal), protected to the north by the monumental earthen Lower Trajan's Wall (*Nizhniy Trayanov Val*, stretching 126 km from the Prut at Vadul-lui-Isac across Lakes Kagul, Yalpug, and Katlabukh to Lake Sasyk) [9][10][42]. Inhabited by mixed Bulgar-Slavic-Vlach agriculturalists and fishermen, Budjak sustained 21,0k inhabitants at 800AD and 25,0k at 900AD (~1,91 persons/km^2, the highest density of any sub-region in the Wild Fields).

Along the Lower Dniester, Southern Bug, and Dnieper Rapids in Sub-regions II, III, and IV (47,0k, 52,0k, and 44,0k at 900AD), East Slavic Tivertsi and Ulichi settlements (Belenkoye, Palanca, Peresechen Steppe, Chern, Voznesensk, Igren-Samar) traded down the rivers toward the Black Sea [19][42]. Simultaneously, in Sub-region V (62,0k at 800 - 900AD) under the *Pax Chazarica*, Alanic and Proto-Bulgar settlers constructed over 250 fortified and open settlements of the Saltovo-Mayaki urban network along the Seversky Donets and Oskol valleys, led by the 120 ha Sidorovo Complex (49,00°N, 37,59°E, 5,0k inhabitants), Mayaki-Tsaritsyno (3,0k), Sukhaya Gomolsha (2,6k), Mokhnach (2,5k), Netailovka (2,2k), Teplinskoye (2,0k), Zlivki (1,4k), and Krivaya Luka (1,3k) [6][23][43][44][45]. Across all five sub-regions, Wild Fields population climbed to 215,0k in 800AD and 230,0k in 900AD.

Pechenegs, Kievan Rus Ports (Oleshye, Protolche, Maurocastro), and the Cuman Apogee (965AD - 1240AD)

Pecheneg migrations and Prince Sviatoslav's 965AD destruction of Khazaria burned the Donets *Saltovo* forts and disrupted open *Dridu* villages in Budjak, lowering regional population to 195,0k at 1000AD (22,0k in Budjak [I], 40,0k in II, 46,0k in III, 42,0k in IV, 45,0k in V) [6][46]. During the eleventh and twelfth centuries (1055 - 1223AD), however, the commercial condominium of Kievan Rus, Byzantium, and the Polovtsian/Cuman (*Kipchak*) confederation lifted regional population to 235,0k in 1100AD and an all-time pre-modern peak of 265,0k in 1200AD (28,0k in Budjak [I], 54,0k in NW Pontic [II], 62,0k in Right-Bank [III], 56,0k in Left-Bank [IV], 65,0k in Middle Donets [V]) [19][46][47].

At the Danube-Dniester mouths in Sub-region I (28,0k at 1200AD), Byzantine, Galician-Rus (*Berladniki*), Vlach, and Cuman trade revived Maurocastro/Belgorod (at ancient Tyras, 3,0k - 3,5k), Kiliia (2,5k), Smil/Izmail (1,8k), Reni (1,0k), and Orlovka/Kartal (1,2k) [10][11]. At the head of the Dnieper estuary in

Sub-region IV, the Kievan Rus sea-port of Oleshye (46,62°N, 32,76°E, 3,5k at 1200AD) served as the maritime staple for Kiev's trade with Constantinople, paired upstream with the Khortytsia horse-ferry and *Brodnik* staging town of Protolche (47,79°N, 35,15°E, 2,2k) and the 25 ha Samara confluence town of Igren-Samar (2,8k) [19][48]. Along the Middle Seversky Donets in Sub-region V, semi-sedentary Cuman-Alanic winter towns flourished at Sharukhan (49,83°N, 36,68°E, 4,5k), Sugrov (3,2k), and Balin (2,8k), while over 2 500 Polovtsian stone statues (*kamennye baby*) and elite kurgans (such as the Chingul Kurgan on the Molochnaya in Sub-region IV) demarcated twelfth- and early thirteenth-century clan pastures [22][46].

The Ulus of Jochi (Golden Horde), Genoese-Moldavian Budjak Ports, Lithuanian Castles, and the 1400AD Cutoff (1240AD - 1400AD)

Mongol Conquest and the Golden Horde and Genoese Urban Revival in Budjak and Veliky Lug (1240AD - 1345AD)

The Mongol campaigns of 1223AD (Battle of the Kalka River in Sub-region V) and 1238 - 1240AD contracted the population of the Wild Fields from 245,0k in 1240AD to 165,0k in 1250AD (18,0k in Budjak [I], 34,0k in II, 38,0k in III, 36,0k in IV, 39,0k in V) [47][48]. Under the western *Ulus of Jochi* (Prince Nogai d. 1299AD, Khan Toqta 1291 - 1312AD, and Khan Uzbek 1313 - 1341AD), however, the region achieved a major commercial and urban resurgence to 210,0k at 1300AD [20][21][49].

Under Nogai and Toqta, Budjak (Sub-region I: 28,0k at 1300AD) became the premier western maritime hub of the Golden Horde. At the Dniester mouth, Aq Qerman (Maurocastro or Cetatea Alba, 46,20°N, 30,35°E) minted its own coinage, hosted Genoese, Armenian, Greek, Jewish, Moldavian, and Tatar merchant quarters, and reached 15,0k - 16,0k inhabitants at its 1300 - 1345AD peak [10][11]. On the Chilia arm of the Danube Delta, the Genoese-notarial port of Kiliia/Lycostomo (45,44°N, 29,26°E, where Antonio di Ponzò's 1360 - 1361AD notarial registers document massive grain, wax, salt, and wine exports to Pera and Genoa) housed 6,5k - 8,0k inhabitants, flanked by Smil/Izmail (3,5k - 4,5k), Reni (1,3k), and the interior Golden Horde camp of Kara-Bunar/Tatarbunary (1,1k) [11]. In the *Veliky Lug* floodplain of the Left-Bank Lower Dnieper (Sub-region IV: 48,0k at 1300AD), Uzbek and Jani Beg built the 300 ha unfortified Islamic city of Velyki Kuchugury (47,68°N, 35,20°E, with a Friday mosque, glazed minaret, subterranean-heated *hammam*, water conduits, and 5,5k - 6,0k inhabitants), paired with Mamai-Surka (1 100+ Golden Horde burials), Konski Vody/Mechetna Mohyla (4,0k), and Oleshye (2,0k), while Italian charts mapped Porto Pisano (Belosaraiskaya, 1,5k) on the Azov coast of Sub-region V [20][21].

The Black Death, Blue Waters, Timur's 1395AD Campaign, and the 1400AD Terminal Benchmark

Between 1346AD and 1400AD, the interior steppes of Sub-regions II, III, IV, and V suffered severe demographic contraction driven by the Black Death (1346 - 1348AD), the Great Troubles of the Golden Horde (1359 - 1380AD), Grand Duke Algirdas's victory at the Battle of Blue Waters (1362AD, on the Syniukha River in Sub-region III), Timur's 1395AD campaign (which systematically razed Kuchugury, Mamai-Surka, Konski Vody, and the Donets settlements), and the Battle of the Vorskla River (1399AD) [21][49]. By contrast, Sub-region I (Budjak) and the coastal ports of Sub-region II exhibited much greater resilience as the Principality of Moldavia (under Petru I Musat and Alexander the Good, 1390s - 1400AD) fortified Cetatea Alba/Aq Qerman (12,0k in 1400AD), Kiliia/Chilia (6,0k), and Smil/Izmail (3,5k), while Grand Duke Vytautas of Lithuania established fortified customs ports at Dashov/Kara-Kerman (Ochakiv, 2,8k in 1400AD), Khadzhibey/Koczubij (Odesa Bay, 2,5k), Sokolets, Balyklei, and Tyagin [11].

At the 1400AD terminal benchmark, the total population of the Wild Fields stood at 155,0k (24,0k in Sub-region I: Budjak, representing 15,5% of the regional total on 5,5% of the land area; 34,0k in

Sub-region II; 34,0k in Sub-region III; 35,0k in Sub-region IV; and 28,0k in Sub-region V).

Quantitative Synthesis, Visual Analytics, and Complete Benchmark Series (10000BC - 1400AD)

Figures 2, 3, and 4 and Table 2 present the complete demographic synthesis for the Wild Fields across all 26 chronological benchmarks from 10000BC to 1400AD across the five natural sub-regions.

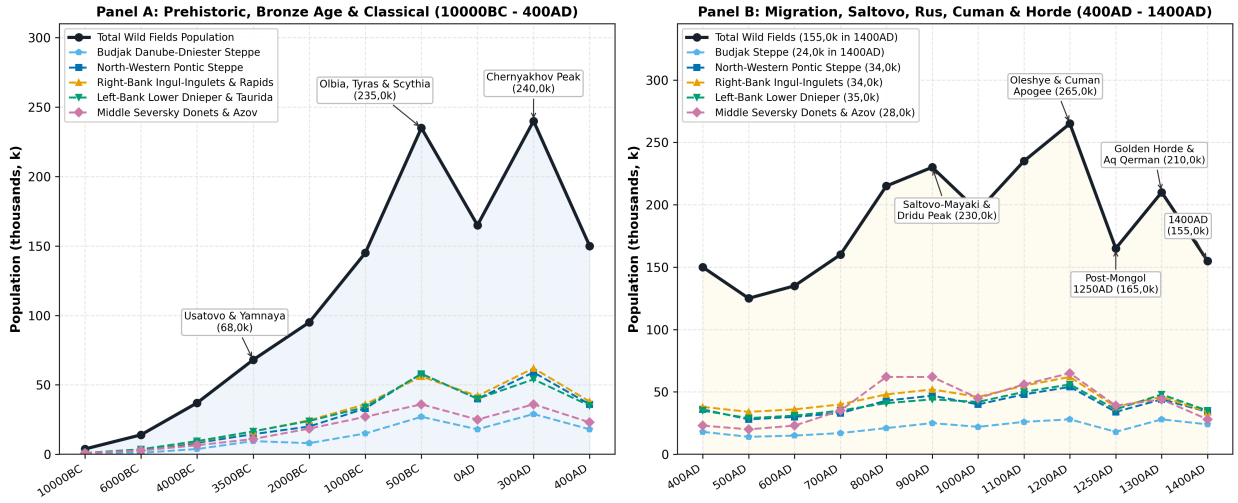


Figure 2. Long-run demographic trajectory of the Wild Fields and Budjak across five natural hydrological sub-regions (10000BC - 1400AD).

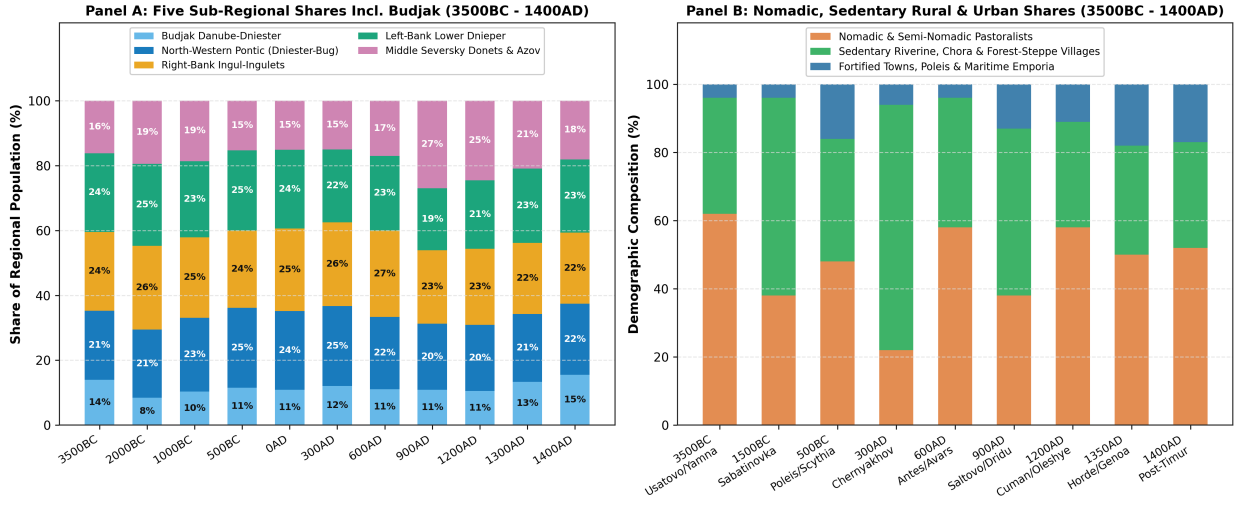


Figure 3. Sub-regional population shares and structural evolution of pastoral, rural, and urban populations in the Wild Fields (3500BC - 1400AD).

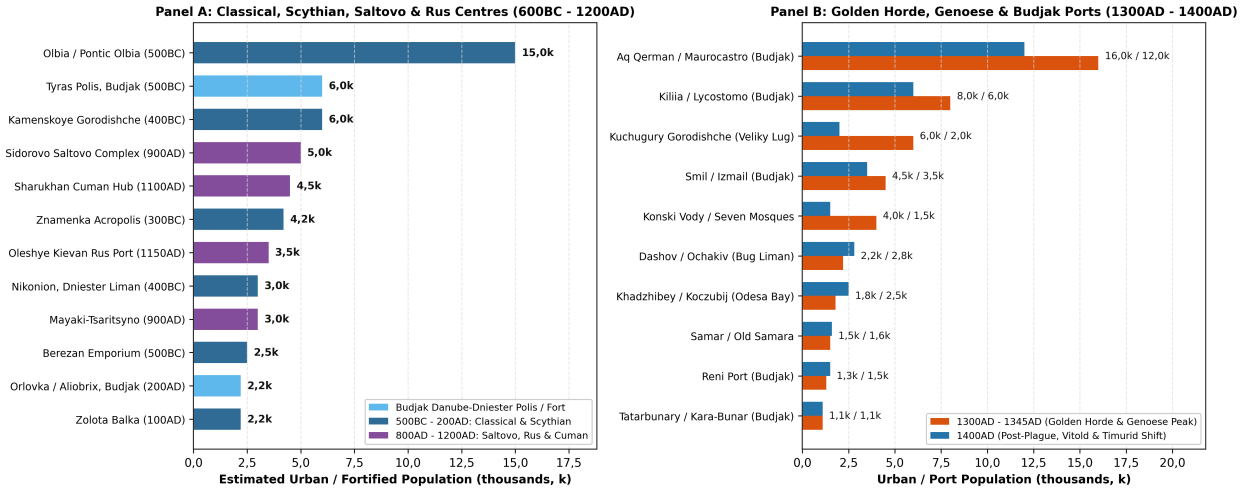


Figure 4. Comparative urban and fortified settlement hierarchies across the Wild Fields and Budjak from classical antiquity to 1400AD.

Table 2. Complete sub-regional and total demographic benchmark series for the Wild Fields and Budjak across five sub-regions (10000BC - 1400AD).

Date	I: Budjak	II: NW Pontic	III: R-Bank	IV: L-Bank	V: Donets	Total	Primary Archaeological / Historical Regime & Anchor Sources
10000BC	0,30k	0,80k	0,95k	0,95k	0,80k	3,8k	Epigravettian bison & horse hunters (Anetovka II, Amvrosievka)
8000BC	0,55k	1,45k	1,7k	1,7k	1,4k	6,8k	Kukrek & Grebeniki Mesolithic camps & rapids cemeteries (Mirnoe [Budjak], Vasylivka)
6000BC	1,1k	3,1k	3,5k	3,6k	2,7k	14,0k	Bug-Dniester, Sursko-Dnieper & Danube-Lakes Early Neolithic villages
5000BC	1,9k	5,3k	6,0k	6,2k	4,6k	24,0k	Dnieper-Donets II & Mariupol Neolithic collective cemeteries (Vovnigi, Mariupol)
4000BC	3,8k	8,2k	9,0k	9,5k	6,5k	37,0k	Bolgrad-Aldeni (Gumelnita) in Budjak; Sredny Stog & Dereivka horse-keepers
3500BC	9,5k	14,5k	16,5k	16,5k	11,0k	68,0k	Budjak Kurgan &

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